

Optimizing Ambulance Deployment Time in Wise County

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Abstract

Wise County spans approximately 405 square miles, making it difficult to ensure full emergency coverage with a limited number of ambulance stations. Due to this geographic spread, it is challenging to reach all areas in an 8-minute response time. The objective of this paper is to minimize the number of ambulances required in key areas of Wise County, including Wise, Norton, and Big Stone Gap, while maintaining a response time of less than 8 minutes. This study develops a mathematical optimization and simulation framework to strategically stage ambulances across six candidate locations. Demand nodes were manually selected based on densely populated areas identified through U.S. Census data. By dynamically relocating ambulances based on time of day and day of week, the model significantly improves system performance. Compared to the current system, the optimized deployment reduces average response times by approximately 53 percent (from 13.4 to 6.3 minutes), demonstrating a more efficient and effective allocation of emergency resources.

1 Introduction

According to the National Center for Biotechnology Information (NCBI), the average Emergency Medical Services (EMS) response time in rural areas is approximately 14.5 minutes, compared to 7.7 minutes in suburban areas and 7.0 minutes in urban areas [2]. This substantial increase in response time can be attributed to several factors, including greater geographic distances, limited funding, and fewer available ambulances. These challenges are especially relevant in Wise County, Virginia, where the combination of mountainous terrain and dispersed population further complicates timely emergency response. In Wise County, the average response time is 13.3 ± 6.86 minutes, highlighting the need for a more efficient deployment strategy.

The primary objective of this study is to minimize ambulance response times while maximizing the proportion of the population that can be reached within 8 minutes, subject to a hard maximum response time of 12 minutes across the service area. To address this, the model focuses on three municipalities: Wise, Norton, and Big Stone Gap. Six candidate staging locations, corresponding to existing emergency facilities, are used as potential ambulance positions. Demand nodes are manually selected based on high-density and high-priority locations, including hospitals, schools, and commercial centers.

To optimize deployment, this study employs a two-layer modeling framework. The first layer uses a coverage-based optimization model to determine the optimal placement and number of ambulances for different time periods. The second layer applies a discrete-event simulation to evaluate these deployment strategies under realistic operating conditions, incorporating factors such as ambulance availability, chute time, and stochastic call arrivals. Together, this approach provides a practical and data-driven solution for improving EMS performance in Wise County.

2 Assumptions

The following assumptions were made to bound the scope of the model:

- All demand nodes are treated with equal weight, regardless of differences in population density or historical call frequency. This reflects uncertainty in precise demand distribution across the service area. Future work will incorporate demand weighting based on historical call frequency to better reflect spatial variability in EMS demand.
- Ambulance travel times are based on the road network and adjusted using time-of-day and day-of-week multipliers derived from historical data. A lights-and-sirens factor of 0.85 is applied uniformly, representing an approximate 15% reduction in travel time under emergency conditions.

- Travel times computed using OpenStreetMap and OSRM are assumed to accurately reflect real-world driving conditions, despite potential variability due to terrain, weather, or road disruptions.
- Each ambulance is assumed to be available at its assigned staging location at the start of each time period. Dynamic repositioning after call completion is not modeled.
- Mutual aid from neighboring jurisdictions is not included. All emergency calls are assumed to be served exclusively by Wise County units.
- Call arrivals follow a Poisson process, with rates derived from historical EMS call volume data for each time period and day of the week.
- Historical EMS data are assumed to be representative of future demand patterns, with no significant structural changes in call behavior.
- Chute time (the delay between dispatch and enroute status) is modeled as a fixed constant (2 minutes) based on supporting literature.
- The six candidate staging locations correspond to existing emergency facilities and remain fixed throughout the study.
- A demand node is considered covered if at least one ambulance can reach it within the response time threshold, regardless of simultaneous call occurrences.

3 Data Processing

For the calculation of time-of-day and day-of-week multipliers, EMS call activity data were obtained from the Wise County Sheriff’s Office covering the past two years [6]. The most relevant fields used in this study included the date of the call, dispatch time (in military format), and the time from dispatch to on-scene arrival. Due to incomplete records, any entries missing one or more of these key variables were excluded from the dataset to ensure accuracy in subsequent analysis.

A total of 47 demand nodes were identified across the three municipalities, prioritizing high-density and high-priority areas such as hospitals, schools, and commercial centers within Wise, Norton, and Big Stone Gap. These locations were selected based on population density data from U.S. Census Dots [3] and facility records from the Virginia Department of Health EMS Communication Centers. Six stationary nodes representing candidate ambulance staging locations were selected manually from existing emergency facilities in the same municipalities.

All nodes, both stationary and demand points, were converted into geographic coordinates for use in the model. For example, the University of Virginia’s College at Wise was represented by the coordinate pair (36.9747, −82.5588). These coordinates enabled distance and travel time calculations between all relevant locations in the study.

4 Distance and Time Estimation

The road network was obtained from OpenStreetMap [5] via the OSMnx Python library [4]. Travel times between all candidate staging locations and demand nodes were computed using OSRM [4], producing a 47×6 base travel-time matrix. A lights-and-sirens factor of 0.85 was applied uniformly to reflect emergency driving conditions.

To capture temporal variation, the base matrix was scaled by time-of-day multipliers derived from Wise County EMS dispatch data [6]. Calls were categorized into five periods: Night (10 pm–6 am), Morning (6 am–10 am),

Midday (10 am–2 pm), Evening (2 pm–7 pm), and Late (7 pm–10 pm). Midday served as the baseline (1.00), with Night at 0.80, Morning at 0.90, Evening at 1.10, and Late at 1.00. Day-of-week multipliers were derived by normalizing average daily call volumes, ranging from 0.86 on Tuesdays to 1.15 on Fridays, and were applied through the simulation layer to scale call arrival rates independently of travel speed.

The adjusted travel time for each origin–destination pair (i, j) under day d and period t is:

$$T_{ij}^{(d,t)} = T_{ij}^{\text{base}} \times 0.85 \times \alpha_t$$

where α_t is the time-of-day multiplier. This produced 35 scenario-specific matrices (7 days $\times$ 5 time periods).

5 Model

5.1 Decision Variables

The Maximal Covering Location Problem (MCLP) model is defined over two sets:

- I : the set of demand nodes (locations where emergency calls may originate)
- J : the set of candidate ambulance staging locations

We define two binary decision variables:

$$\begin{aligned} x_j &\in \{0, 1\} && \text{for each } j \in J \\ y_i &\in \{0, 1\} && \text{for each } i \in I \end{aligned}$$

- $x_j = 1$ if an ambulance is stationed at location j , and 0 otherwise. This represents the deployment decision.
- $y_i = 1$ if demand node i is covered within the response time threshold $T = 8$ minutes by at least one ambulance, and 0 otherwise.

We also define the coverage parameter a_{ij} :

$$a_{ij} = \begin{cases} 1 & \text{if travel time from location } j \text{ to node } i \leq 8 \text{ minutes} \\ 0 & \text{otherwise} \end{cases}$$

The parameter a_{ij} is precomputed from the travel time matrix for each of the 35 day/period scenarios.

5.2 Objective Function

$$\max \sum_{i \in I} y_i$$

The objective is to maximize the total number of demand nodes covered within the 8-minute response time threshold. Each node contributes equally, reflecting the goal of equitable emergency response access across the service area.

5.3 Constraints

Coverage constraint:

$$y_i \leq \sum_{j \in J} a_{ij} x_j \quad \forall i \in I$$

A demand node i can only be considered covered if at least one selected staging location j can reach it within the threshold time. If no such location exists, then $y_i = 0$.

Fleet size constraint:

$$\sum_{j \in J} x_j \leq p$$

The total number of active staging locations cannot exceed the available fleet size p . The model is solved iteratively for $p = 1, 2, 3, \dots$ until the coverage target of 90% is achieved.

Binary constraints:

$$\begin{aligned} x_j &\in \{0, 1\} & \forall j \in J \\ y_i &\in \{0, 1\} & \forall i \in I \end{aligned}$$

Both decision variables are binary, making this a binary integer programming model.

6 Results

6.1 Optimization Results

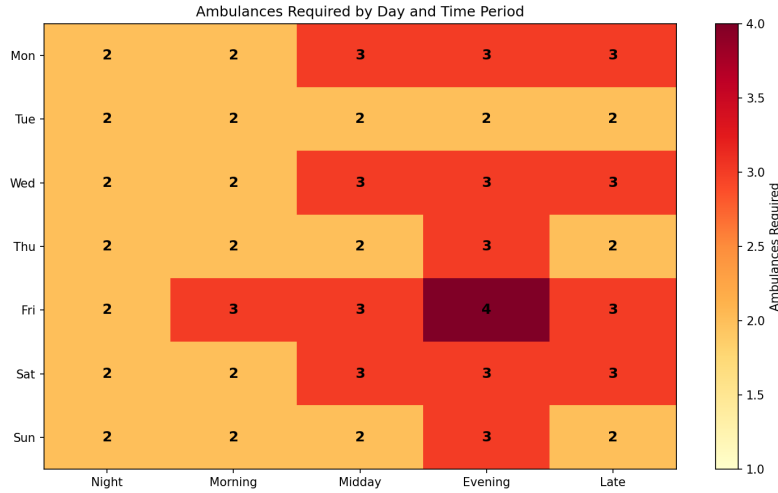


Figure 1: Ambulances required by day and time period under optimized staging.

The Maximum Covering Location Problem (MCLP) was solved for all 35 day-period combinations, producing the recommended fleet size and staging locations for each scenario. As shown in Figure 1, the required fleet size ranges from 2 ambulances during low-demand periods to 4 ambulances during Friday evening peak hours, with

the majority of periods requiring 3 units. Tuesday is the lightest day, requiring only 2 ambulances throughout, while Friday is the most demanding, peaking at 4 units in the evening. The recommended operational fleet size is therefore 4 ambulances.

6.2 Response Time Performance

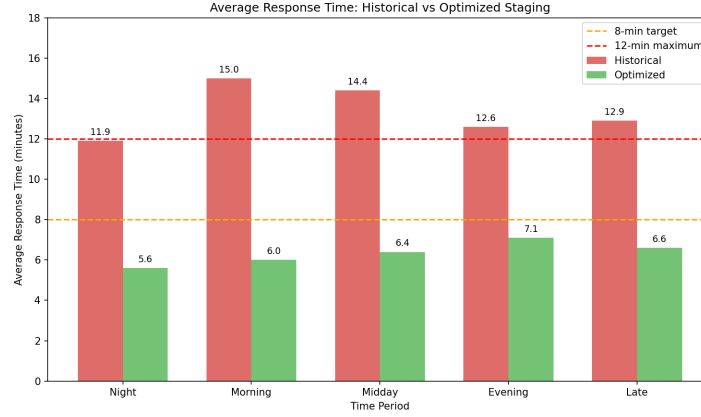


Figure 2: Average response time by time period: historical vs. optimized staging.

Figure 2 compares average response times between the historical system and the optimized staging plan across all five time periods. The optimized strategy reduces average response time from a historical range of 11.9–15.0 minutes down to 5.6–7.1 minutes across all periods. The largest improvement occurs during morning hours, where mean response time drops from 15.0 to 6.0 minutes. Evening remains the most challenging period with a mean of 7.1 minutes under the optimized plan, though this still represents a 5.5-minute improvement over the historical 12.6-minute average.

6.3 Coverage Performance

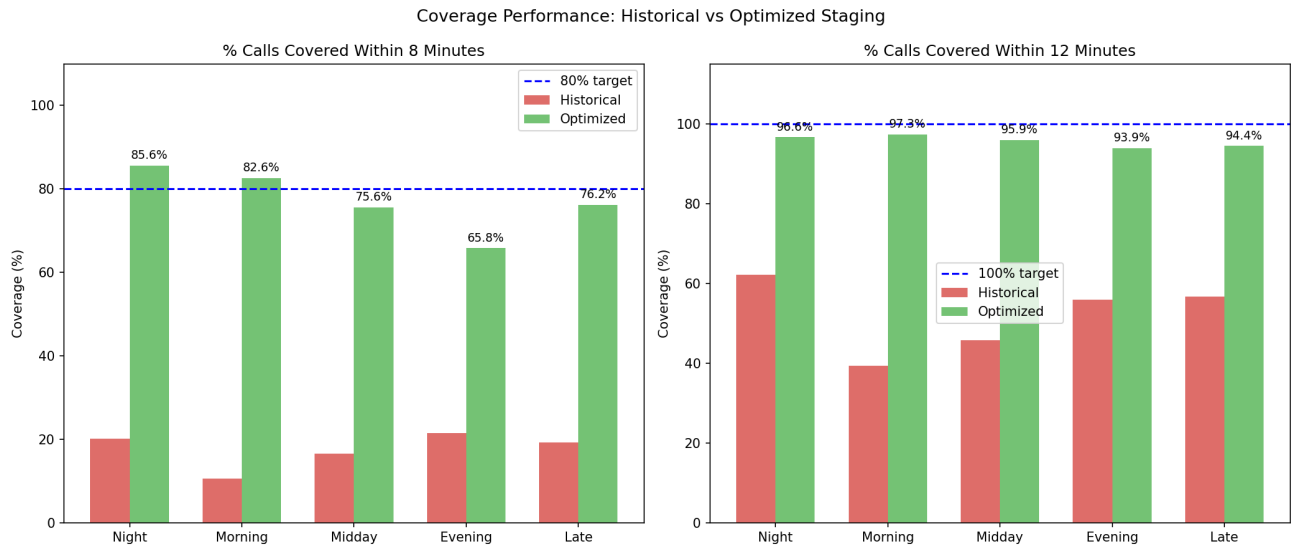


Figure 3: Coverage performance within 8 and 12 minutes for historical and optimized systems.

Figure 3 shows coverage percentages within 8 and 12 minutes for both the historical and optimized systems. Under the optimized plan, 8-minute coverage ranges from 65.8% during evening to 85.6% at night, a substantial improvement over the historical range of 10.6–21.5%. All time periods exceed the 80% coverage target except for evening (65.8%), which falls short due to the combined effect of peak traffic multipliers and high call volume. Coverage within 12 minutes reaches 93.9–97.3% across all periods, compared to 39.4–62.2% historically.

6.4 Ambulance Staging Schedule



Figure 4: Recommended ambulance staging locations by day and time period.

Figure 4 presents the recommended ambulance staging schedule across all 35 day-period combinations. Each cell identifies the specific stations to activate and the number of ambulances required to achieve the coverage target for that scenario. Color intensity encodes fleet size: light blue for 2 units, medium blue for 3, and dark blue for the Friday evening peak of 4. The schedule reveals two dominant configurations. The two-ambulance configuration, staging at Wise Fire Department and Lonesome Pine Hospital, covers the geographic extremes of the service area and is sufficient for all overnight and low-demand periods. The three-ambulance configuration adds Norton Fire Department and BSG Fire Department, distributing coverage across all three municipalities during higher-demand periods. Tuesday is the only day that remains on the two-unit configuration throughout all five periods, consistent with it being the lightest call volume day of the week. Friday evening is the sole scenario requiring all four ambulances, driven by the combined effect of peak traffic multipliers and the highest day-of-week demand weight. This schedule is intended as a practical operational reference for Wise County EMS dispatchers, providing clear shift-by-shift guidance on where to position available units throughout the week.

6.5 Response Time Distribution

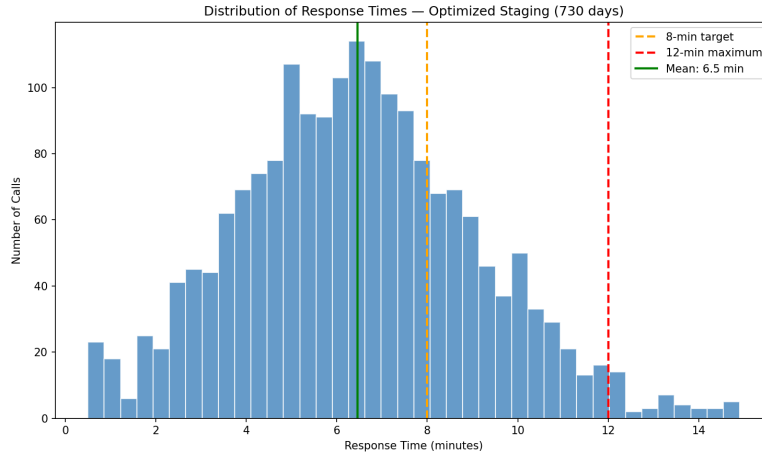


Figure 5: Distribution of response times under optimized staging across all simulated calls.

Figure 5 shows the full distribution of response times across all 1,874 simulated calls. The distribution is right-skewed, with the majority of calls concentrated between 3 and 9 minutes. The mean response time of 6.3 minutes falls below the 8-minute target line, indicating that the typical call is well-served under the optimized plan. The tail of the distribution extends to approximately 24 minutes, representing a small proportion of calls from remote locations or during periods of simultaneous ambulance deployment.

6.6 City-Level Performance

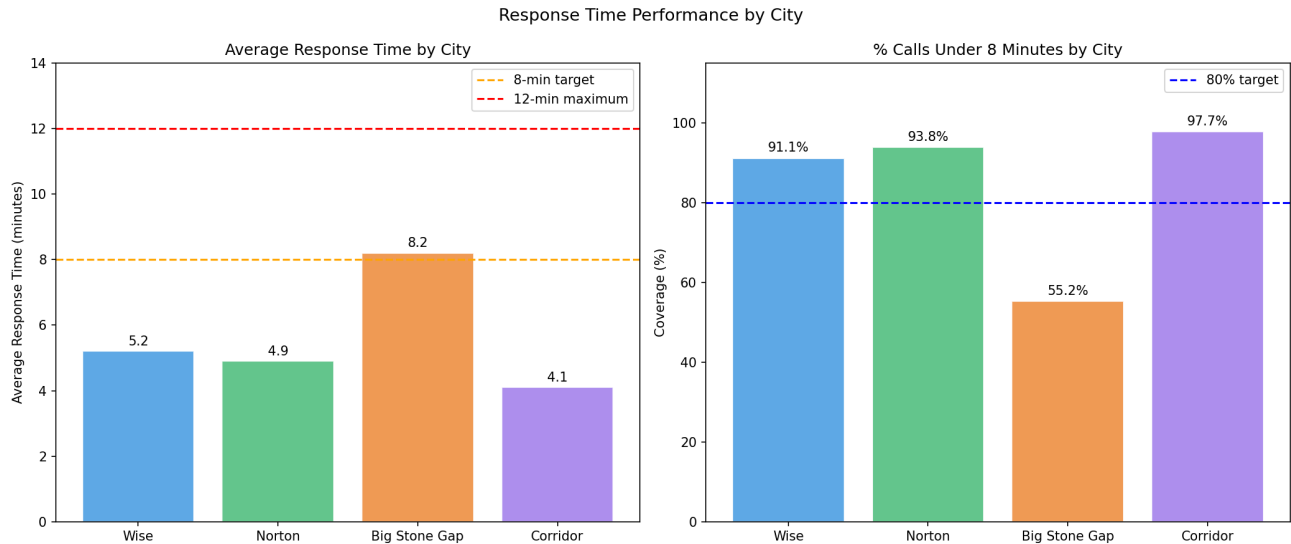


Figure 6: Average response time and coverage under 8 minutes by city.

Figure 6 reveals significant geographic disparity in response time performance across the three communities. Norton achieves the best performance with a mean response time of 4.9 minutes and 93.8% of calls under 8 minutes. Wise follows closely at 5.2 minutes and 91.1% coverage. Big Stone Gap is the most challenging area with a mean

response time of 8.2 minutes and only 55.2% of calls under 8 minutes, reflecting its geographic isolation at the southwestern extreme of the service area. The corridor nodes between towns achieve the fastest response times at 4.1 minutes on average, benefiting from proximity to multiple staging locations.

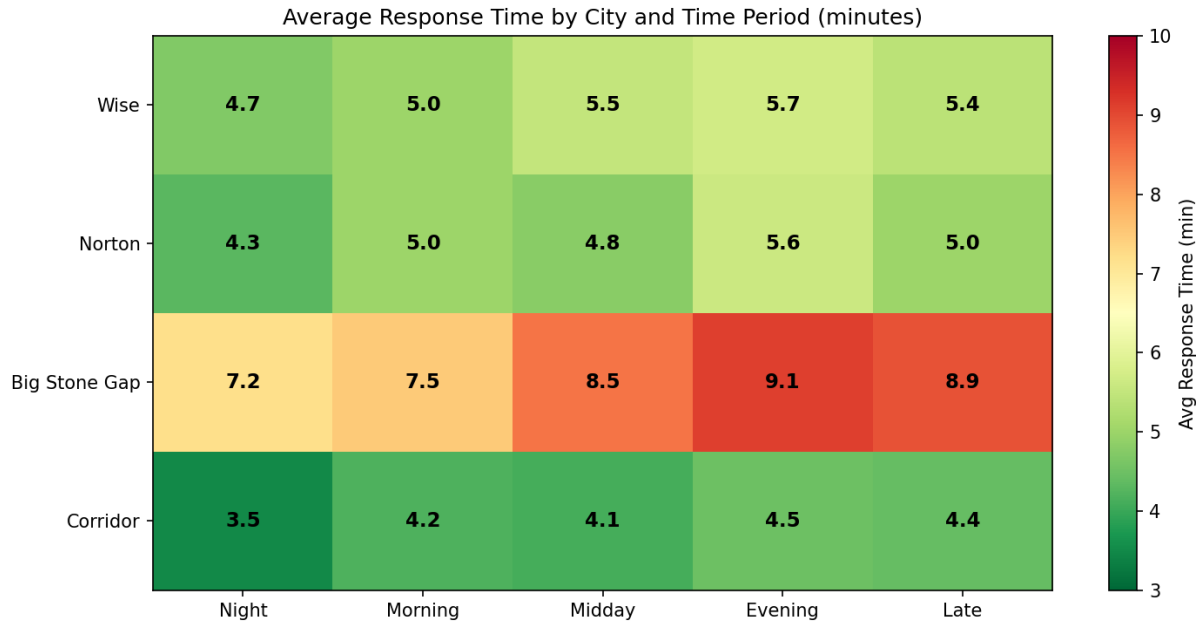


Figure 7: Average response time by city and time period.

Figure 7 provides a granular breakdown of average response time by city and time period. The contrast between Big Stone Gap and the other communities is consistent across all periods, with Big Stone Gap ranging from 7.2 minutes at night to 9.1 minutes during evening. Norton and Wise remain well below the 8-minute threshold across all periods, while corridor nodes are consistently the fastest.

6.7 Outlier Analysis

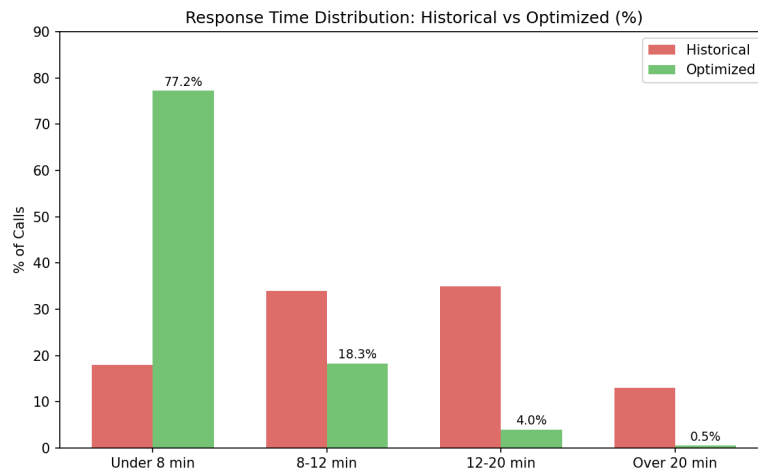


Figure 8: Response time distribution comparison: historical vs. optimized staging.

Figure 8 compares the response time distribution between historical and optimized systems across four response time bands. Under the optimized plan, 77.2% of calls are answered within 8 minutes compared to approximately 18% historically. The proportion of calls exceeding 12 minutes drops from approximately 48% historically to 4.5% under the optimized plan, and no calls exceed 30 minutes. The remaining 4.5% of calls exceeding 12 minutes represent remote geographic locations or rare simultaneous call scenarios rather than systemic failures in the staging strategy.

7 Discussion

7.1 Key Findings

The results demonstrate that dynamic ambulance staging produces substantial improvements in emergency response performance across all time periods and all three communities. The core finding is that a fleet of just 4 ambulances, strategically positioned according to a day-of-week and time-of-day schedule, can achieve average response times of 6.3 minutes and 77.2% coverage within 8 minutes, compared to 13.4 minutes and approximately 18% under the current system. This represents a 53% reduction in average response time.

The two-station night configuration, Wise Fire Department and Lonesome Pine Hospital, is particularly notable. By positioning one unit at each geographic extreme of the service area, the model achieves 85.6% night coverage with only 2 ambulances. This is operationally significant because reduced overnight staffing is a real constraint for rural EMS systems, and the model confirms that high coverage is achievable even with a minimal fleet during low-demand hours.

7.2 Ambulance Availability

Across 730 simulated days and 1,874 calls, no call experienced a complete absence of available ambulances. This indicates that the recommended fleet size is sufficient to handle simultaneous calls without leaving any demand node entirely unserved. The 4.5% of calls exceeding 12 minutes are attributable to geographic distance rather than ambulance unavailability, as confirmed by the zero instances of calls with no ambulance available.

7.3 Limitations

The model has several limitations that should be considered when interpreting results. First, demand nodes were assumed to have equal weight, which does not account for population density differences between locations. A population-weighted MCLP would more accurately reflect true call probability across the service area. Second, the staging plan uses fixed time blocks rather than continuous redeployment, meaning an ambulance clearing a call is not dynamically repositioned; it simply becomes available at its last location. Third, mutual aid from neighboring jurisdictions is not modeled, which means the worst-case response times represent scenarios where Wise County units alone must respond. In practice, mutual aid agreements would provide additional coverage during peak demand or simultaneous call scenarios. Finally, the simulation assumes a fixed staging configuration for each time period, whereas real operations would involve dynamic repositioning as ambulances clear calls throughout the day.

7.4 Future Work

Several extensions would strengthen this model. Adding population weights to demand nodes would improve the realism of the MCLP objective. Incorporating dynamic redeployment, repositioning available ambulances after clearing calls, would reduce response times beyond what the static staging plan achieves. Collecting detailed

on-scene and transport time data from Wise County records would replace the literature-based estimates currently used in the simulation. Finally, exploring additional staging locations along the US-23 corridor between Norton and Big Stone Gap could address the persistent coverage gap in that area and improve Big Stone Gap response times.

8 Conclusion

This study presents a data-driven approach to optimizing ambulance deployment in Wise County, Virginia, through the use of a coverage-based optimization model combined with discrete-event simulation. By leveraging real EMS call data, OpenStreetMap-based travel times, and population-informed demand nodes, the model provides a realistic representation of emergency response conditions in a rural setting.

The results show that a strategically deployed fleet of 4 ambulances is sufficient to achieve high coverage across the majority of time periods, with 8-minute coverage ranging from 65.8% during Friday evening to 85.6% at night, and 12-minute coverage reaching 93.9–97.3% across all periods. Compared to the current system, the optimized deployment reduces average response times from 13.4 minutes to 6.3 minutes and increases 8-minute coverage from approximately 18% to 77.2%. These improvements demonstrate a significant increase in both efficiency and accessibility of emergency care.

The model also reveals key insights into spatial deployment, particularly the importance of maintaining coverage at geographic extremes during low-demand periods and shifting toward more distributed coverage during peak times. While a small number of remote locations remain outside the 12-minute threshold due to geographic constraints, overall system performance is substantially improved.

This framework demonstrates that meaningful improvements in EMS performance can be achieved not by increasing fleet size, but through more efficient and adaptive deployment strategies. Future work could incorporate real-time data, dynamic repositioning, and probabilistic availability models to further enhance system responsiveness. Overall, this study highlights the value of combining real-world data with mathematical optimization to address critical challenges in rural emergency medical services.

References

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